Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/30/2012
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	In vivo induction of chromosome instability and aberrant patterns of DNA methylation in hematopoietic stem/progenitor cells by heavy ions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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Flight Program:			
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COI Name (Institution):	Dunn, John (Brookhaven Science Associates/Dept of Energy) Whorton, Elbert Ph.D. (Galveston National Laboratory)		
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Task Description:	Space environment consists of various types of radiation that are different from those found in the Earth's atmosphere. These include particles with high mass and high energy (heavy particles) such as silicon and titanium. It has been estimated that a manned mission to Mars may take longer than 500 days. During a journey to Mars, every cell in the body may be hit by these particles several times within a month. Hence, exposure to radiation in space is the greatest hazard to astronauts venturing beyond Earth. It is therefore important to protect astronauts in space environments. This will require improvement in our knowledge of radiation-induced changes in specific target cells that may impact the health of astronauts. Cancer of blood cells, in particular myeloid leukemia, is one of the major health concerns from exposure to space radiation. However, the predictions of risks for developing cancer of blood cells after exposure to space radiation are unsatisfactory. To increase the reliability of predicting risk for developing cancer of blood cells, a much improved understanding of space radiation-induced changes in the target cells (stem cells and their progenitors) in living organisms for the induction of this particular type of cancer is critically important. In this project, we will expose mice to space-radiation with different qualities or linear energy transfer (LET) selected from the range of heavy ions found in space and relevant to NASA missions, i.e. silicon and titanium. We will determine the effects of these ions on the induction of damage at the genetic level by measuring the frequency and type of initial and late-occurring chromosome instability (normally known as genomic instability). Such analyses will be done in hematopoietic stem/progenitor cells (HSPCs) residing in the bone marrow (BM) of exposed mice collected at different times post-irradiation using a novel genome-wide multicolor fluorescence in situ hybridization (whole genome mFISH). The measurement of chromosome instability in HSPCs will be coupled with the characterization of damage at the epigenetic level (i.e. the damage affecting gene function without a change in the sequence of DNA molecules) by determining the pattern of DNA-methylation. These two biological endpoints are highly relevant for assessing cancer risks; however, they have not been applied with in vivo exposure to space radiation with different LET. Hence, the results will provide high-priority information for NASA. Our research is the first to use integrative in vivo endpoints in the HSPCs that are the target of radiation-induced myeloid leukemia to evaluate the LET effects of space-radiation. Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve our understanding of the molecular basis for radiation dose and LET dependencies of genomic instability and aberrant DNA-methylation profiles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of the in vivo work proposed in this study, our data will provide pivotal information that will lead to a better understanding of the patterns of radiation-induced genetic and non-genetic (epigenetic) damage in HSPCs that trigger the induction of genomic instability, a key event of carcinogenesis. This study will couple molecular mechanisms with quantitative data from a robust and highly sensitive technique. The resulting data will deliver information that is of high priority for NASA because the data will not only contribute to reducing the uncertainties in the assessment of cancer risk from exposure to radiation (heavy ions) during space flights but will also improve the understanding of the molecular basis for radiation-induced genomic instability. Hence, the obtained data are critically important for understanding risks associated with space travel as well as developing advanced radiation (heavy ions) therapies for cancer treatment on Earth. Moreover, the information obtained from this project will have a significant impact on the future development of a novel mechanism-based therapeutic strategy to prevent or mitigate the acute or long-term effects of radiation on cells at risk for myeloid leukemia induced by radiation found in space, as well as those found on Earth.
Task Progress:	This project was initiated in July 2011. Therefore, this first task progress represents our accomplishments of the past 10 months. The major goals during the first year are to: 1. complete the exposure of mice to different doses of 28Si ions, 2. collect BM cells from sham controls and exposed mice, 3. obtain colonies of BM-derived HSPCs at different time up to six months after irradiation, 4. determine the number and types (according to morphology) of the BM-derived HSPCs colonies after 7 days incubation, 5. collect BM-derived HSPCs obtained from sham controls and exposed mice for the analyses of chromosome aberrations (CAs) and DNA methylation induced in vivo, 6. measure the frequency of CAs and determine the pattern of DNA methylation in HSPCs obtained from sham controls and exposed mice. We have accomplished goals 1 to 4. With respect to goal #6, the measurements of the frequency of CAs by means of the whole genome mFISH in metaphase cells prepared from HSPC colonies derived from BM cells collected at all harvest times (1 wk, 1 mo, and 6 mos) after in vivo exposure to 28Si ions are underway. Metaphases have been prepared, kept in fixative at 4oC, and are readily available for slide preparation. Although the mFISH method is robust and capable of measuring the frequency, the degree of complexity, and the dose response for the production and persistence of simple and complex aberrations over time, it is very time consuming. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide CAs by mFISH is our current major effort and will be continued into year 2 of the study. During the next years, studies will focus on the characterization of the type and distribution of the initial and the subsequent loss of chromosomal damage, and the expression of genomic instability induced in vivo in BM cells by 28Si ions. Dose and time response relationships will be established for the induction of genomic instability. The results will aid in determining whether or not there are unique biological effects associated with exposure to 28Si ions. Relating to the determination of the pattern of DNA methylation in HSPCs, we are exploring several new techniques that will enable the selective detection, capture, and enrichment of 5-hydroymethylcytosine (5hmC) genomic DNA fragments for mapping this interesting epigenetic mark at single-nucleotide resolution. The most suitable methods will be adapted for high-throughput sequence data acquisition using BNL’s newly acquired Ion Torrent instrument. The analysis of the DNA methylation (both 5mC and 5hmC) will be the major efforts in the following years of studies.

Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Tungjai M, Honikel, L, Rithidech K. "Comparative effects of 28Si ions on the heart and the bone marrow tissues of whole-body irradiated mice." 23rd Annual NASA Space Radiation Health Investigators' Workshop, Durham, North Carolina, July 8-July 11, 2012. Presented at the Radiation-induced tissue damage forum. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, North Carolina, July 8-July 11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Rithidech K, Dunn JJ, Whorton EB. "In vivo induction of chromosome instability and aberrant patterns of DNA methylation in heamotopoietic stem/progenitor cells (HSPCs) by 28Si ions." 23rd Annual NASA Space Radiation Health Investigators' Workshop, Durham, North Carolina, July 8-July 11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, North Carolina, July 8-July 11, 2012. , Jul-2012